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United States Patent	12410138
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Schotes; Christoph et al.

Process comprising the use of new iridium catalysts for enantioselective hydrogenation of 4-substituted 1,2-dihydroquinolines

Abstract

New iridium catalysts for enantioselective hydrogenation of 4-substituted 1,2-dihydroquinolines The invention relates to a process for preparing optically active 4-substituted 1,2,3,4-tetra hydroquinolines (la, lb) comprising enantioselective hydrogenation of the corresponding 4-substituted 1,2-dihydroquinolines in presence of a chiral iridium (P,N)-ligand catalyst. ##STR00001##

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Appl. No.:	**17/640287**
Filed (or PCT Filed):	**September 22, 2020**
PCT No.:	**PCT/EP2020/076375**
PCT Pub. No.:	**WO2021/058458**
PCT Pub. Date:	April 01, 2021

Prior Publication Data

Document Identifier	**Publication Date**
US 20220324810 A1	Oct. 13, 2022

Foreign Application Priority Data

EP	19199636	Sep. 25, 2019

Publication Classification

Int. Cl.: **C07D215/08** (20060101)

U.S. Cl.:

CPC **C07D215/08** (20130101);

Field of Classification Search

CPC: C07D (215/08)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is the National Stage entry of International Application No. PCT/EP2020/076375, filed 22 Sep. 2020, which claims priority to European Patent Application No. 19199636.2, filed 25 Sep. 2019. Each of these applications is incorporated by reference in its entirety.
BACKGROUND
Field
(2) The invention relates to a process for preparing optically active 4-substituted 1,2,3,4-tetrahydroquinolines comprising enantioselective hydrogenation of the corresponding 4-substituted 1,2-dihydroquinolines in presence of a chiral iridium (P,N)-ligand catalyst.
Description of Related Art
(3) It is known from EP 0 654 464 that N-acetyl-tetrahydroquinolines can be converted to the corresponding 4-aminoindane derivatives via a rearrangement reaction. 4-aminoindane derivatives are important intermediates for preparing various N-indanyl heteroaryl carboxamides having fungicidal activity (EP 0 654 464, WO 2011/162397, WO 2012/084812, WO 2015/197530).
(4) EP 3 103 789 discloses a method for optically resolving 1,1,3-trimethyl-4-aminoindane by converting the enantiomeric mixture into the diastereomeric salts of D-tartaric acid. (R)- and (S)-1,1,3-trimethyl-4-aminoindane are obtained after separation and basification of the diastereomeric salts. This reference also discloses a method for racemizing the undesired enantiomer, so that the whole method allows for converting the undesired enantiomer into the desired enantiomer via several process steps. (R)-1,1,3-trimethyl-4-aminoindane is an important intermediate for preparing the pyrazole carboxamide fungicide inpyrfluxam.
(5) A method for preparing chiral intermediates of N-indanyl heteroaryl carboxamides via

asymmetic synthesis is also known. WO 2015/141564 describes a process for preparing optically active 4-substituted 1,2,3,4-tetrahydroquinolines, which process comprises the hydrogenation of the corresponding 4-substituted 1,2-dihydroquinolines in presence of a transition metal catalyst having an optically active ligand. The asymmetric hydrogenation of the 4-substituted NH-dihydroquinolines proceeded with moderate conversion rates (up to 62.6%) and enantioselectivity (up to 71.3% ee), whereas N-acetyl-dihydroquinolines gave even poorer conversion (up to 14%) and enantioselectivity (up to 31% ee).

SUMMARY

(6) In the light of the prior art described above, it is an object of the present invention to provide a process for preparing optically active 4-substituted 1,2,3,4-tetrahydroquinolines which process has advantages over the processes of the prior art. The process should allow the desired enantiomer to be prepared in high yield and high enantiomeric purity, with few process steps and few purification steps.

(7) The object described above was achieved by a process for preparing a compound of the formula (Ta) or (Tb),

(8) ##STR00002## wherein R.sup.1 is selected from the group consisting of C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.1-C.sub.6-alkoxy-C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl, C.sub.6-C.sub.14-aryl, or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein the C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl and the C.sub.1-C.sub.6-alkoxy in the C.sub.1-C.sub.6-alkoxy-C.sub.1-C.sub.6-alkyl moiety, are optionally substituted by 1 to 3 substituents independently selected from the group consisting of halogen, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-haloalkoxy and phenyl, wherein the phenyl may be substituted by one to five substituents selected independently from each other from halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkyl, and C.sub.1-C.sub.4-haloalkoxy, and wherein the C.sub.6-C.sub.14-aryl and the C.sub.6-C.sub.14-aryl in the C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl moiety in each case is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, R.sup.2 and R.sup.3 are the same and are selected from the group consisting of hydrogen, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl and C.sub.1-C.sub.6-alkoxy-C.sub.1-C.sub.6-alkyl, or R.sup.2 and R.sup.3 together with the carbon which they are bound to, form a C.sub.3-C.sub.6-cycloalkyl ring, R.sup.4 is hydrogen, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.1-C.sub.6-alkoxy, C.sub.1-C.sub.6-haloalkoxy, C.sub.1-C.sub.6-alkylamino, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, C.sub.3-C.sub.6-cycloalkyl, C.sub.3-C.sub.6-cycloalkyl-C.sub.1-C.sub.4-alkyl, C.sub.2-C.sub.6-alkenyloxy, 9-flurorenylmethyleneoxy, C.sub.6-C.sub.14-aryl, C.sub.6-C.sub.14-aryloxy, C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyloxy or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein the C.sub.6-C.sub.14-aryl as such or as part of a composite substituent is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, n is 0, 1, 2, 3 or 4, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.1-C.sub.6-alkoxy, hydroxyl, amino and —C(═O)—C.sub.1-C.sub.6-alkyl, comprising enantioselective hydrogenation of a compound of the formula (II)

(9) ##STR00003## wherein the substituents R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5 and the integer n are each as defined for the compound of the formula (Ta) or (Tb), in presence of a chiral iridium catalyst, characterized in that the chiral iridium catalyst comprises a chiral ligand of the formula (IIIa) or (IIIb),

(10) ##STR00004## wherein R.sup.6 is a group of formula

(11) ##STR00005## wherein ** denotes the bond to the 6,7-dihydro-5H-cyclopenta[b]pyridine moiety, R.sup.13 is hydrogen, methyl or ethyl, R.sup.14 is C.sub.1-C.sub.6-alkyl, R.sup.7 is

hydrogen, R.sup.8 is C.sub.1-C.sub.4-alkyl or phenyl, wherein the phenyl is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, R.sup.9 and R.sup.10 are independently from one another selected from the group consisting of C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.5-cycloalkyl, piperidinyl and pyridyl, or R.sup.9 and R.sup.10 together with the phosphorus atom which they are bound to form a group G.sup.1 or G.sup.2
(12) ##STR00006## in which the bonds identified by “x” and “y” are both bound directly to the phosphorus atom, p and q are independently from one another selected from 0, 1 and 2, R.sup.11 and R.sup.12 are independently selected from C.sub.1-C.sub.6-alkyl and phenyl, which may be substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-alkoxy and phenyl, which may be substituted by one or two C.sub.1-C.sub.4-alkyl substituents, or R.sup.9 and R.sup.10 together with the phosphorus atom which they are bound to form a group G.sup.3
(13) ##STR00007## in which the bonds identified by “u” and “v” are both bound directly to the phosphorus atom.

Description

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT
(1) It has been found, surprisingly, that optically active 4-substituted 1,2,3,4-tetrahydroquinolines (Ia and Ib) can be prepared in high yields and excellent enantioselectivity by enantioselective hydrogenation of the corresponding 4-substituted 1,2-dihydroquinolines (II) in presence of a chiral iridium (P,N)-ligand catalyst.
Definitions
(2) In the definitions of the symbols given in the above formulae, collective terms were used, which are generally representative of the following substituents:
(3) Halogen: fluorine, chlorine, bromine or iodine, preferably fluorine, chlorine or bromine, and more preferably fluorine or chlorine.
(4) Alkyl: saturated, straight-chain or branched hydrocarbyl substituents having 1 to 6, preferably 1 to 4 carbon atoms, for example (but not limited to) C.sub.1-C.sub.6-alkyl such as methyl, ethyl, propyl (n-propyl), 1-methylethyl (iso-propyl), butyl (n-butyl), 1-methylpropyl (sec-butyl), 2-methylpropyl (iso-butyl), 1,1-dimethylethyl (tert-butyl), pentyl, 1-methylbutyl, 2-methylbutyl, 3-methylbutyl, 2,2-dimethylpropyl, 1-ethylpropyl, 1,1-dimethylpropyl, 1,2-dimethylpropyl, hexyl, 1-methylpentyl, 2-methylpentyl, 3-methylpentyl, 4-methylpentyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 1,3-dimethylbutyl, 2,2-dimethylbutyl, 2,3-dimethylbutyl, 3,3-dimethylbutyl, 1-ethylbutyl, 2-ethylbutyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1-ethyl-1-methylpropyl and 1-ethyl-2-methylpropyl. Particularly, said group is a C.sub.1-C.sub.4-alkyl group, e.g. a methyl, ethyl, propyl, 1-methylethyl (isopropyl), butyl, 1-methylpropyl (sec-butyl), 2-methylpropyl (iso-butyl) or 1,1-dimethylethyl (tert-butyl) group. This definition also applies to alkyl as part of a composite substituent, for example C.sub.3-C.sub.6-cycloalkyl-C.sub.1-C.sub.4-alkyl, C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl etc., unless defined elsewhere.
(5) Alkenyl: unsaturated, straight-chain or branched hydrocarbyl substituents having 2 to 6, preferably 2 to 4 carbon atoms and one double bond in any position, for example (but not limited to) C.sub.2-C.sub.6-alkenyl such as vinyl, allyl, (E)-2-methylvinyl, (Z)-2-methylvinyl, isopropenyl, homoallyl, (E)-but-2-enyl, (Z)-but-2-enyl, (E)-but-1-enyl, (Z)-but-1-enyl, 2-methylprop-2-enyl, 1-methylprop-2-enyl, 2-methylprop-1-enyl, (E)-1-methylprop-1-enyl, (Z)-1-methylprop-1-enyl, pent-4-enyl, (E)-pent-3-enyl, (Z)-pent-3-enyl, (E)-pent-2-enyl, (Z)-pent-2-enyl, (E)-pent-1-enyl, (Z)-pent-1-enyl, 3-methylbut-3-enyl, 2-methylbut-3-enyl, 1-methylbut-3-enyl, 3-methylbut-2-enyl,

(E)-2-methylbut-2-enyl, (Z)-2-methylbut-2-enyl, (E)-1-methylbut-2-enyl, (Z)-1-methylbut-2-enyl, (E)-3-methylbut-1-enyl, (Z)-3-methylbut-1-enyl, (E)-2-methylbut-1-enyl, (Z)-2-methylbut-1-enyl, (E)-1-methylbut-1-enyl, (Z)-1-methylbut-1-enyl, 1,1-dimethylprop-2-enyl, 1-ethylprop-1-enyl, 1-propylvinyl, 1-isopropylvinyl, (E)-3,3-dimethylprop-1-enyl, (Z)-3,3-dimethylprop-1-enyl, hex-5-enyl, (E)-hex-4-enyl, (Z)-hex-4-enyl, (E)-hex-3-enyl, (Z)-hex-3-enyl, (E)-hex-2-enyl, (Z)-hex-2-enyl, (E)-hex-1-enyl, (Z)-hex-1-enyl, 4-methylpent-4-enyl, 3-methylpent-4-enyl, 2-methylpent-4-enyl, 1-methylpent-4-enyl, 4-methylpent-3-enyl, (E)-3-methylpent-3-enyl, (Z)-3-methylpent-3-enyl, (E)-2-methylpent-3-enyl, (Z)-2-methylpent-3-enyl, (E)-1-methylpent-3-enyl, (Z)-1-methylpent-3-enyl, (E)-4-methylpent-2-enyl, (Z)-4-methylpent-2-enyl, (E)-3-methylpent-2-enyl, (Z)-3-methylpent-2-enyl, (E)-2-methylpent-2-enyl, (Z)-2-methylpent-2-enyl, (E)-1-methylpent-2-enyl, (Z)-1-methylpent-2-enyl, (E)-4-methylpent-1-enyl, (Z)-4-methylpent-1-enyl, (E)-3-methylpent-1-enyl, (Z)-3-methylpent-1-enyl, (E)-2-methylpent-1-enyl, (Z)-2-methylpent-1-enyl, (E)-1-methylpent-1-enyl, (Z)-1-methylpent-1-enyl, 3-ethylbut-3-enyl, 2-ethylbut-3-enyl, 1-ethylbut-3-enyl, (E)-3-ethylbut-2-enyl, (Z)-3-ethylbut-2-enyl, (E)-2-ethylbut-2-enyl, (Z)-2-ethylbut-2-enyl, (E)-1-ethylbut-2-enyl, (Z)-1-ethylbut-2-enyl, (E)-3-ethylbut-1-enyl, (Z)-3-ethylbut-1-enyl, 2-ethylbut-1-enyl, (E)-1-ethylbut-1-enyl, (Z)-1-ethylbut-1-enyl, 2-propylprop-2-enyl, 1-propylprop-2-enyl, 2-isopropylprop-2-enyl, 1-isopropylprop-2-enyl, (E)-2-propylprop-1-enyl, (Z)-2-propylprop-1-enyl, (E)-1-propylprop-1-enyl, (Z)-1-propylprop-1-enyl, (E)-2-isopropylprop-1-enyl, (Z)-2-isopropylprop-1-enyl, (E)-1-isopropylprop-1-enyl, (Z)-1-isopropylprop-1-enyl, 1-(1,1-dimethylethyl)ethenyl, buta-1,3-dienyl, penta-1,4-dienyl, hexa-1,5-dienyl or methylhexadienyl. Particularly, said group is vinyl or allyl. This definition also applies to alkenyl as part of a composite substituent unless defined elsewhere.
(6) Alkynyl: straight-chain or branched hydrocarbyl substituents having 2 to 8, preferably 2 to 6, and more preferably 2 to 4 carbon atoms and one triple bond in any position, for example (but not limited to) C.sub.2-C.sub.6-alkynyl, such as ethynyl, prop-1-ynyl, prop-2-ynyl, but-1-ynyl, but-2-ynyl, but-3-ynyl, 1-methylprop-2-ynyl, pent-1-ynyl, pent-2-ynyl, pent-3-ynyl, pent-4-ynyl, 2-methylbut-3-ynyl, 1-methylbut-3-ynyl, 1-methylbut-2-ynyl, 3-methylbut-1-ynyl, 1-ethylprop-2-ynyl, hex-1-ynyl, hex-2-ynyl, hex-3-ynyl, hex-4-ynyl, hex-5-ynyl, 3-methylpent-4-ynyl, 2-methylpent-4-ynyl, 1-methylpent-4-ynyl, 2-methylpent-3-ynyl, 1-methylpent-3-ynyl, 4-methylpent-2-ynyl, 1-methylpent-2-ynyl, 4-methylpent-1-ynyl, 3-methylpent-1-ynyl, 2-ethylbut-3-ynyl, 1-ethylbut-3-ynyl, 1-ethylbut-2-ynyl, 1-propylprop-2-ynyl, 1-isopropylprop-2-ynyl, 2,2-dimethylbut-3-ynyl, 1,1-dimethylbut-3-ynyl, 1,1-dimethylbut-2-ynyl, or 3,3-dimethylbut-1-ynyl group. Particularly, said alkynyl group is ethynyl, prop-1-ynyl, or prop-2-ynyl. This definition also applies to alkynyl as part of a composite substituent unless defined elsewhere.
(7) Alkylamino: monoalkylamino or dialkylamino, wherein monoalkylamino represents an amino radical having one alkyl residue with 1 to 4 carbon atoms attached to the nitrogen atom. Non-limiting examples include methylamino, ethylamino, n-propylamino, isopropylamino, n-butylamino and tert-butylamino. Wherein dialkylamino represents an amino radical having two independently selected alkyl residues with 1 to 4 carbon atoms each attached to the nitrogen atom. Non-limiting examples include N,N-dimethylamino, N,N-diethyl-amino, N,N-diisopropylamino, N-ethyl-N-methylamino, N-methyl-N-n-propylamino, N-isopropyl-N-n-propylamino and N-tert-butyl-N-methylamino.
(8) Alkoxy: saturated, straight-chain or branched alkoxy substituents having 1 to 6, more preferably 1 to 4 carbon atoms, for example (but not limited to) C.sub.1-C.sub.6-alkoxy such as methoxy, ethoxy, propoxy, 1-methylethoxy, butoxy, 1-methylpropoxy, 2-methylpropoxy, 1,1-dimethylethoxy, pentoxy, 1-methylbutoxy, 2-methylbutoxy, 3-methylbutoxy, 2,2-dimethylpropoxy, 1-ethylpropoxy, 1,1-dimethylpropoxy, 1,2-dimethylpropoxy, hexoxy, 1-methylpentoxy, 2-methylpentoxy, 3-methylpentoxy, 4-methylpentoxy, 1,1-dimethylbutoxy, 1,2-dimethylbutoxy, 1,3-dimethylbutoxy, 2,2-dimethylbutoxy, 2,3-dimethylbutoxy, 3,3-dimethylbutoxy, 1-ethylbutoxy, 2-ethylbutoxy, 1,1,2-trimethylpropoxy, 1,2,2-trimethylpropoxy, 1-ethyl-1-methylpropoxy and 1-ethyl-2-methylpropoxy.

This definition also applies to alkoxy as part of a composite substituent unless defined elsewhere.
(9) Cycloalkyl: mono- or polycyclic, saturated hydrocarbyl substituents having 3 to 12, preferably 3 to 8 and more preferably 3 to 6 carbon ring members, for example (but not limited to) cyclopropyl, cyclopentyl, cyclohexyl and adamantyl. This definition also applies to cycloalkyl as part of a composite substituent, for example C.sub.3-C.sub.6-cycloalkyl-C.sub.1-C.sub.4-alkyl, unless defined elsewhere.
(10) Haloalkyl: straight-chain or branched alkyl substituents having 1 to 6, preferably 1 to 4 carbon atoms (as specified above), where some or all of the hydrogen atoms in these groups are replaced by halogen atoms as specified above, for example (but not limited to) C.sub.1-C.sub.3-haloalkyl such as chloromethyl, bromomethyl, dichloromethyl, trichloromethyl, fluoromethyl, difluoromethyl, trifluoromethyl, chlorofluoromethyl, dichlorofluoromethyl, chlorodifluoromethyl, 1-chloroethyl, 1-bromoethyl, 1-fluoroethyl, 2-fluoroethyl, 2,2-difluoroethyl, 2,2,2-trifluoroethyl, 2-chloro-2-fluoroethyl, 2-chloro-2,2-difluoroethyl, 2,2-dichloro-2-fluoroethyl, 2,2,2-trichloroethyl, pentafluoroethyl and 1,1,1-trifluoroprop-2-yl. This definition also applies to haloalkyl as part of a composite substituent unless defined elsewhere.
(11) Haloalkenyl and haloalkynyl are defined analogously to haloalkyl except that, instead of alkyl groups, alkenyl and alkynyl groups are present as part of the substituent.
(12) Haloalkoxy: straight-chain or branched alkoxy substituents having 1 to 6, preferably 1 to 4 carbon atoms (as specified above), where some or all of the hydrogen atoms in these groups are replaced by halogen atoms as specified above, for example (but not limited to) C.sub.1-C.sub.3-haloalkoxy such as chloromethoxy, bromomethoxy, dichloromethoxy, trichloromethoxy, fluoromethoxy, difluoromethoxy, trifluoromethoxy, chlorofluoromethoxy, dichlorofluoromethoxy, chlorodifluoromethoxy, 1-chloroethoxy, 1-bromoethoxy, 1-fluoroethoxy, 2-fluoroethoxy, 2,2-difluoroethoxy, 2,2,2-trifluoroethoxy, 2-chloro-2-fluoroethoxy, 2-chloro-2,2-difluoroethoxy, 2,2-dichloro-2-fluoroethoxy, 2,2,2-trichloroethoxy, pentafluoroethoxy and 1,1,1-trifluoroprop-2-oxy. This definition also applies to haloalkoxy as part of a composite substituent, unless defined elsewhere.
(13) Aryl: mono-, bi- or tricyclic aromatic or partially aromatic substituents having 6 to 14 carbon atoms, for example (but not limited to) phenyl, naphthyl, tetrahydronapthyl, indenyl and indanyl. The binding to the superordinate general structure can be carried out via any possible ring member of the aryl residue. Aryl is preferably selected from phenyl, 1-naphthyl, 2-naphthyl, 9-phenantryl und 9-antracenyl. Phenyl is particularly preferred.
(14) The term “enantioselective” as used herein means that one of the two possible enantiomers of the hydrogenation product, namely the enantiomer of the formula (Ia) or the enantiomer of the formula (Ib), is preferably formed. The “enantiomeric excess” or “ee” indicates the degree of enantioselectivity:
(15) $\%ee = \frac{\text{major enantiomer (mol) - minor enantiomer (mol)}}{\text{major enantiomer (mol) + minor enantiomer (mol)}} \times 100\%$
(16) The major enantiomer can be controlled by the selection of the chiral ligand, for example by selecting the chiral ligand of the formula (IIIa) or the opposite enantiomer (the ligand of the formula (IIIb)).
(17) The process according to the invention is used for preparing the compound of the formula (Ta) or (Tb), preferably (Ia).
(18) Preferred are compounds of the formula (Ia) or (Ib), in particular (Ia), wherein the substituents are defined as follows: R.sup.1 is C.sub.1-C.sub.6-alkyl or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein C.sub.6-C.sub.14-aryl in the C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl moiety is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkoxy, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is

independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl.
(19) More preferred are compounds of the formula (Ia) or (Ib), in particular (Ia), wherein the substituents are defined as follows: R.sup.1 is C.sub.1-C.sub.6-alkyl R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, or R.sup.2 and R.sup.3 together with the carbon which they are bound to, form a C.sub.3-C.sub.6-cycloalkyl ring, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen and C.sub.1-C.sub.6-alkyl.
(20) Even more preferred are compounds of the formula (Ia) or (Ib), in particular (Ia), wherein the substituents are defined as follows: R.sup.1 is methyl, ethyl or n-propyl, R.sup.2 and R.sup.3 are methyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen and C.sub.1-C.sub.6-alkyl.
(21) Most preferred are compounds of the formula (Ia) or (Ib), in particular (Ia), wherein the substituents are defined as follows: R.sup.1 is methyl or n-propyl, R.sup.2 and R.sup.3 are methyl, R.sup.4 is methyl, n is 0 or 1, substituent R.sup.5, if present, is fluorine.
(22) The process according to the invention comprises enantioselective hydrogenation of the compound of the formula (II). The substituents R.sup.1, R.sup.2, R, R.sup.4, R.sup.5 and the integer n in the compound of the formula (II) are each as defined for the compound of the formula (Ta) or (Tb).
(23) The enantioselective hydrogenation of the compound of the formula (II) is conducted in presence of a chiral iridium catalyst comprising a chiral ligand of the formula (IIIa) or (IIIb).
(24) In a preferred embodiment of the process according to the invention, the substituents of formulae (Ia), (Ib), (II), (IIIa), (IIIb) are defined as follows: R.sup.1 is C.sub.1-C.sub.6-alkyl or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein C.sub.6-C.sub.14-aryl in the C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl moiety is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkoxy, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl, R.sup.6 is a group of formula
(25) ##STR00008## wherein ** denotes the bond to the 6,7-dihydro-5H-cyclopenta[b]pyridine moiety, R.sup.13 is hydrogen, methyl or ethyl, R.sup.14 is C.sub.1-C.sub.4 alkyl, R.sup.7 is hydrogen, R.sup.8 is C.sub.1-C.sub.4 alkyl or phenyl, wherein the phenyl is unsubstituted or substituted by one to five C.sub.1-C.sub.4-alkyl substituents, R.sup.9 and R.sup.10 are independently from one another selected from the group consisting of iso-propyl, tert-butyl, cyclopentyl, cyclohexyl and piperidin-1-yl.
(26) In a more preferred embodiment of the process according to the invention, the substituents of formulae (Ia), (Ib), (II), (IIIa), (IIIb) are defined as follows: R.sup.1 is C.sub.1-C.sub.6-alkyl, R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkoxy, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl, R.sup.6 is a group of formula
(27) ##STR00009## wherein ** denotes the bond to the 6,7-dihydro-5H-cyclopenta[b]pyridine moiety, R.sup.13 is ethyl, R.sup.14 is methyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are independently from one another selected from the group consisting of cyclohexyl and piperidin-1-yl.
(28) In the most preferred embodiment of the process according to the invention, the substituents of formulae (Ia), (Ib), (II), (IIIa), (IIIb), are defined as follows: R.sup.1 is C.sub.1-C.sub.4-alkyl,

R.sup.2 and R.sup.3 are methyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, n is 0 or 1, R.sup.5 if present, is fluorine, R.sup.6 is 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl.
(29) Depending on whether compound (Ia) or (Ib) is the desired product, the ligand of the formula (IIIa) or (IIIb) is selected.
(30) Preferred are ligands of the formulae (IIIa) and (IIIb), wherein the substituents are defined as follows: R.sup.6 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl.
(31) Preferably, the chiral iridium catalyst is selected from the group consisting of [IrL*(COD)]Y and [IrL*(nbd)]Y, wherein L* is the chiral ligand of the formulas (IIIa) and (IIIb), COD represents 1,5-cyclooctadiene, nbd represents norbornadiene, and Y is a non-coordinating anion selected from the group consisting of [B(R.sup.18).sub.4].sup.−, PF.sub.6.sup.− and [Al{OC(CF.sub.3).sub.3}.sub.4].sup.− of formula (VII)
(32) ##STR00010## wherein R.sup.18 is selected from fluorine and phenyl, which is unsubstituted or substituted with one to five substituents selected from C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl and halogen.
(33) More preferred are chiral iridium catalysts of the formulae [IrL*(COD)]Y and [IrL*(nbd)]Y, wherein Y is [Al{OC(CF.sub.3)3}.sub.4].sup.− of formula (VII) or [B(R.sup.18).sub.4].sup.−, wherein R.sup.18 is phenyl, which is unsubstituted or substituted with one to five substituents selected from fluorine and trifluoromethyl.
(34) Even more preferred are chiral iridium catalysts of the general formulae (Va) and (Vb)
(35) ##STR00011## wherein R.sup.6 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl Y is a non-coordinating anion selected from the group consisting of [B(R.sup.18).sub.4].sup.− and [Al{OC(CF.sub.3).sub.3}.sub.4].sup.− of formula (VII), wherein R.sup.18 is 3,5-bis(trifluoromethyl)phenyl.
(36) Most preferred are chiral iridium catalysts of the general formula (Va)
(37) ##STR00012## wherein R.sup.6 is 2,4,6-trimethylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl, Y is [Al{OC(CF.sub.3)3}.sub.4].sup.− of formula (VII).
(38) The amount of iridium catalyst used is preferably within the range of from 0.001 mol % to 5 mol %, more preferably 0.001 mol % to 4 mol %, most preferably 0.002 mol % to 3 mol %, in particular 0.005 mol % to 1.0 mol %, based on the amount of the compound of the formula (II).
(39) The chiral iridium catalyst may be prepared by methods known in the art from an iridium (I) catalyst precursor, such as [Ir(COD)Cl].sub.2, the chiral ligand of the formula (IIIa) or (IIIb) and an alkali salt of the non-coordinating anion (S. Kaiser et al., Angew. Chem. Int. Ed. 2006, 45, 5194-5197; W. J. Drury III et al., Angew. Chem. Int. Ed. 2004, 43, 70-74).
(40) The process according to the invention comprises enantioselective hydrogenation of the compound of the formula (II).
(41) Preferably, the hydrogenation is conducted using hydrogen gas at a pressure of from 1 to 300 bar, preferably 3 to 200 bar, most preferably 20 to 150 bar.
(42) The hydrogenation is preferably conducted at a temperature within the range of from 20° C. to 130° C., more preferably 30° C. to 100° C.
(43) Suitable solvents are halogenated alcohols such as 2,2,2,-trifluoroethanol, hexafluoroisopropanol (1,1,1,3,3,3-hexafluoro-2-propanol) and tetrafluoropropanol (2,2,3,3-tetrafluoro-1-propanol), halogenated hydrocarbons, such as chlorobenzene, dichlorobenzene, dichloromethane, chloroform, tetrachloromethane, dichloroethane and trichloroethane, aromatic hydrocarbons such as benzene, toluene and xylene, ethers such as diethyl ether, diisopropyl ether, methyl tert-butyl ether, methyl tert-amyl ether, dioxane, tetrahydrofuran, 1,2-dimethoxyethane, 1,2-diethoxyethane and anisole, and esters such as ethyl acetate, isopropyl acetate, and mixtures thereof.

(44) Preferred solvents are selected from the group consisting of 2,2,2,-trifluoroethanol, hexafluoroisopropanol, 1,2-dichloroethane, tetrafluoropropanol, 1,4-dioxane, isopropyl acetate, toluene, and mixtures thereof.
(45) More preferred solvents are selected from the group consisting of 2,2,2,-trifluoroethanol, hexafluoroisopropanol, 1,2-dichloroethane, tetrafluoropropanol, and mixtures thereof.
(46) Especially preferred are 2,2,2,-trifluoroethanol and hexafluoroisopropanol.
(47) Most preferred is hexafluoroisopropanol.
(48) The process according to the invention may optionally be conducted in the presence of an additive, which is selected from the group consisting of Bronsted acids and Lewis acids.
(49) In a preferred embodiment of the process according to the invention, the additive is selected from the group consisting of hexafluorophosphoric acid, acetic acid, trifluoromethylsulfonic acid, water, pentafluorophenol, 3,5-bis(trifluoromethyl)phenol, tetrafluoroboric acid, tetrafluoroboric acid diethylether complex, nafion, amberlyst, 1,1,1,3,3,3-hexafluoro-2-(trifluoromethyl)propan-2-ol, triphenylborane, tris[3,5-bis(trifluoro-methyl)phenyl]borane, tris(2,3,4,5,6-pentafluorophenyl)borane, borane tetrahydrofurane complex, boric acid, aluminum (III) trifluoromethanesulfonate, zinc (II) trifluoromethanesulfonate, scandium (III) trifluoro-methanesulfonate, aluminum (III) fluoride, titanium (IV) isopropoxide, trimethyl aluminum, boron trifluoride, complexes of boron trifluoride, and mixtures thereof.
(50) Suitable complexes of boron trifluoride are complexes of boron trifluoride with organic solvents, such as dialkyl ethers or alcohols, and complexes of boron trifluoride with organic acids, such as carboxylic acids. Preferred boron trifluoride complexes are selected from the group consisting of boron trifluoride-diethylether complex, boron trifluoride acetic acid complex and boron trifluoride n-propanol complex.
(51) In a more preferred embodiment of the process according to the invention, the additive is selected from the group consisting of hexafluorophosphoric acid, pentafluorophenol, 3,5-bis(trifluoromethyl)phenol, tetra-fluoroboric acid diethylether complex, triphenylborane, tris[3,5-bis(trifluoromethyl)phenyl]borane, tris(2,3,4,5,6-pentafluorophenyl)borane, aluminum (III) trifluoromethanesulfonate, scandium (III) trifluoro-methanesulfonate, aluminum (III) fluoride, titanium (IV) isopropoxide, trimethyl aluminum, boron trifluoride, complexes of boron trifluoride, and mixtures thereof, wherein the complexes of boron trifluoride are preferably selected from the group consisting of boron trifluoride-diethylether complex, boron trifluoride acetic acid complex and boron trifluoride n-propanol complex.
(52) In a even more preferred embodiment of the process according to the invention, the additive is selected from the group consisting of hexafluorophosphoric acid, pentafluorophenol, 3,5-bis(trifluoromethyl)phenol, tri-phenylborane, tris[3,5-bis(trifluoromethyl)phenyl]borane, tris(2,3,4,5,6-pentafluorophenyl)borane, aluminum (III) trifluoromethanesulfonate, scandium (III) trifluoromethanesulfonate, aluminum (III) fluoride, titanium (IV) isopropoxide, trimethyl aluminum, boron trifluoride, complexes of boron trifluoride, and mixtures thereof, wherein the complexes of boron trifluoride are preferably selected from the group consisting of boron trifluoride-diethylether complex, boron trifluoride acetic acid complex and boron trifluoride n-propanol complex.
(53) In the most preferred embodiment of the process according to the invention, the additive is selected from the group consisting Most preferred are aluminum (III) trifluoromethanesulfonate, scandium (III) trifluoro-methanesulfonate, tris(2,3,4,5,6-pentafluorophenyl)borane, hexafluorophosphoric acid, boron trifluoride and complexes of boron trifluoride, wherein the complexes of boron trifluoride are preferably selected from the group consisting of boron trifluoride diethylether complex, boron trifluoride acetic acid complex and boron trifluoride n-propanol complex.
(54) The amount of additive selected from the group consisting of Bronsted acids and Lewis acids used is preferably within the range of from 0.1 mol % to 10 mol %, more preferably 0.2 mol % to 5

mol %, most preferably 0.3 mol % to 2 mol %, in particular 0.4 mol % to 1 mol %, based on the amount of the compound of the formula (II).

Abbreviations and Acronyms

(55) TABLE-US-00001 a/a Ac Acetyl BARF Tetrakis[3,5-bis(trifluoromethyl)phenyl]-borate CgPBr 8-bromo-1,3,5,7-tetramethyl-2,4,6-trioxa-8- phosphatricyclo[3.3.1.13,7]decane c-hexane cyclohexane ClP-(S-BINOL) (S)-1,1′-Binaphthyl-2,2′-diyl phosphorochloridate Cy Cyclohexyl DCM dichloromethane GC-FID Gas chromatography - Flame ionization detector HPLC High performance liquid chromatography Et Ethyl Me Methyl n-BuLi n-Butyllithium PTFE Polytetrafluoroethylene RT Room temperature SFC Supercritical fluid chromatography THF tetrahydrofurane Tf Trifluoromethylsulfonyl TFE 2,2,2-Trifluoroethanol

Preparation of Iridium Catalysts

(56) ##STR00013##

(57) The ligand precursors (enantiomerically enriched secondary alcohols) were prepared according to known literature procedures like to the method disclosed in S. Kaiser et al., Angew. Chem. Int. Ed. 2006, 45, 5194-5197 or in D. H. Woodmansee Chem. Sci 2010, 1, 72. The ligands and iridium complexes were prepared by a modified procedure based on the same literature precedents:

(58) Standard Procedures

(59) Procedure of ligand synthesis (under Ar): A solution of alcohol precursor in THF (0.25 mmol, in 5.0 mL THF) was cooled to −78° C. and n-BuLi (0.1 mL of a 2.5 M n-BuLi solution in hexane; 0.25 mmol; 1 eq.) was added dropwise to the continuously stirred solution. After completion of the addition the solution was allowed to warm to room temperature and was stirred at this temperature for further 30 min. The solution was cooled to −78° C. again and $R^9R^{10}PCl$ (0.25 mmol, 1 eq.) was added to the continuously stirred solution. The mixture was allowed to warm to room temperature and subsequently heated to 50° C. and kept at this temperature overnight. The theoretical yield of ligand was calculated using ^{31}P-NMR and the ligand was used for the next step without further purification.

(60) Procedure of complexation (under Ar): To the crude ligand solution was added $[Ir(COD)_2]BARF$ (BARF=Tetrakis[3,5-bis(trifluoromethyl)phenyl]-borate) (as a solid, 1 eq. based on the theoretical yield). The resulting mixture was heated to 50° C. and kept at this temperature for 3 h.

(61) Work-up (under air): After cooling to room temperature the reaction solution is rotary evaporated onto silica, loaded onto a column of silica. Side components were eluted using pentane/diethylether and the desired complexes subsequently with DCM. The solvent was then evaporated under reduced pressure.

(62) The following specified catalysts were synthesized and characterized:

(63) ##STR00014##

(64) TABLE-US-00002 TABLE 1 Catalyst R^8 R^9, R^{10} R^7 R^{13} R^{14} $[Y]^-$ Va-1 methyl Cyclohexyl H Et Me Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-2 methyl cyclohexyl H Et Me $[Al\{OC(CF_3)_3\}_4]^-$ of formula (VII) Va-3 phenyl cyclohexyl H Et Me Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-4 methyl S-BINOL*-yl H Et Me Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-5 methyl piperidin-1-yl H Et Me Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-6 methyl cyclohexyl H Me tBu PF_6^- Va-7 methyl Cg-Phos** H Me H Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-8 methyl tBu, 2-pyridyl H Me H Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-9 methyl cyclohexyl H Me H $[Al\{OC(CF_3)_3\}_4]^-$ of formula (VII) Va-10 methyl cyclohexyl H Me Me Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-11 methyl cyclohexyl H Me tBu Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-12 methyl cyclohexyl H Me H Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate Va-13 H phenyl H H H Tetrakis(3,5- bis(trifluoromethyl)phenyl)borate BINOL* (S)-(—)-1,1′-Binaphthalene-2,2′-diol,

embedded image
Va-1
(65) The reaction was performed according to the above described standard procedure. The complex could be isolated as an orange solid (282 mg; 76% based on [Ir(COD).sub.2]BARF).
(66) .sup.1H-NMR (300 MHz, CD.sub.2Cl.sub.2): δ (ppm)=7.73 (s, 8H), 7.56 (s, 4H), 7.23 (s, 1H), 7.14 (s, 1H), 7.00 (s, 1H), 5.62 (dd, J=8.0, 5.6 Hz, 1H), 5.46-5.39 (m, 1H), 4.32 (dd, J=7.4, 3.4 Hz, 1H), 3.36-3.27 (m, 1H), 3.19-3.06 (m, 2H), 3.01-2.91 (m, 2H), 2.80 (dq, J=14.9, 7.4 Hz, 1H), 2.59-2.43 (m, 2H), 2.43-2.15 (m, 7H), 2.15-0.83 (m, 36H), 0.66-0.48 (m, 1H). .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ (ppm)=119.00. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ (ppm)=−62.87. HR-MS (ESI) m/z calcd for C40H58NOPIr [M].sup.+ 792.3880 found 792.3903.
(67) Va-2
(68) A solution of the respective alcohol precursor in THF (0.25 mmol, in 5.0 mL THF) was cooled to −78° C. and n-BuLi (0.1 mL of a 2.5 M n-BuLi solution in hexane; 0.25 mmol; 1 eq.) was added dropwise to the continuously stirred solution. After completion of the addition the solution was allowed to warm to room temperature and was stirred at this temperature for further 30 min. The solution was cooled to −78° C. again and Cy.sub.2PCl (0.25 mmol, 1 eq.) was added to the continuously stirred solution. The mixture was allowed to warm to room temperature and subsequently heated to 50° C. and kept at this temperature overnight. After the reaction was cooled down to RT, THF was removed and dried in vacuum, [Ir(COD)Cl].sub.2 (0.125 mmol) and DCM (5.0 mL) were added to the tube, stirred at 50° C. for 2 h. Then Li{A1[OC(CF.sub.3).sub.3].sub.4} (0.275 mmol) was added to the reaction mixture and stirred at RT for overnight. The reaction solution is rotary evaporated onto silica, loaded onto a column of silica prepared with DCM chromatographed with n-heptane/DCM: 1/1 to afford the orange solid (140 mg, 32%).
(69) .sup.1H-NMR (300 MHz, CD.sub.2Cl.sub.2) δ=7.27 (s, 1H), 7.18 (s, 1H), 7.04 (s, 1H), 5.66 (dt, J=8.0, 3.8 Hz, 1H), 5.50-5.44 (m, 1H), 4.40-4.31 (m, 1H), 3.46-3.26 (m, 1H), 3.25-2.91 (m, 4H), 2.91-2.77 (m, 1H), 2.57-2.52 (m, 2H), 2.50-2.20 (m, 7H), 2.19-1.78 (m, 13H), 1.70-1.53 (m, 5H), 1.49-1.02 (m, 18H), 0.62 (s, 1H). .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=119.02. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ=−75.74. HR-MS (ESI) m/z calcd for C.sub.45H.sub.60NOPIr [M].sup.+ 792.3880 found 792.3903.
(70) Va-3
(71) The reaction was performed according to the above described standard procedure. The complex could be isolated as an orange solid (162 mg; 42% based on [Ir(COD).sub.2]BARF).
(72) .sup.1H-NMR (300 MHz, CD.sub.2Cl.sub.2) δ=7.80-7.65 (m, 8H), 7.65-7.41 (m, 10H), 7.17 (s, 1H), 7.04 (s, 1H), 5.70-5.65 (m, 1H), 5.53-5.47 (m, 1H), 4.42-4.35 (m, 1H), 3.43-3.32 (m, 2H), 3.27-3.09 (m, 2H), 3.09-2.93 (m, 1H), 2.86 (dq, J=14.8, 7.4 Hz, 1H), 2.55-2.48 (m, 2H), 2.38 (s, 3H), 2.27-1.46 (m, 19H), 1.46-0.74 (m, 18H), 0.69-0.52 (m, 1H). .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=119.07. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ=−62.87. HR-MS (ESI) m/z calcd for C.sub.45H.sub.60NOPIr [M].sup.+ 854.4036 found 854.4073.
(73) Va-4
(74) The reaction was performed using ClP-(S)-BINOL (0.25 mmol, freshly prepared) and [Ir(COD).sub.2]BARF (287 mg, 0.225 mmol) according to the above described standard procedure. The complex could be isolated as an orange solid (104 mg) after three times column chromatography using Heptane/DCM: 1/1. There are still four peaks showing in .sup.31P NMR, the main peak is in 117.72 ppm (around 80% purity). The HRMS shows that the desired complex is present.
(75) .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=120.36, 117.72, 114.61, 92.27. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ=−62.85. HR-MS (ESI) m/z calcd for C.sub.48H.sub.48NO.sub.3PIr [M].sup.+ 910.2996 found 910.3026.
(76) Va-5
(77) The reaction was performed according to the above described standard procedure, using 1,1′-

(chlorophosphanediyl)dipiperidine (0.25 mmol, freshly prepared). The complex could be isolated as a red solid (149 mg). There are 6 peaks showing in .sup.31P NMR, the main peak is in 102.35 and 98.87 ppm (2:3). The HRMS shows that the desired complex is present.
(78) .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=102.35, 101.96, 101.40, 98.87, 98.46, 98.35. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ=−62.87. HR-MS (ESI) m/z calcd for C.sub.39H.sub.58N.sub.3OPIr [M].sup.+ 808.3941 found 808.3958.
(79) Va-6
(80) A solution of the respective alcohol precursor in THF (0.25 mmol, in 5.0 mL THF) was cooled to −78° C. and n-BuLi (0.1 mL of a 2.5 M n-BuLi solution in hexane; 0.25 mmol; 1 eq.) was added dropwise to the continuously stirred solution. After completion of the addition the solution was allowed to warm to room temperature and was stirred at this temperature for further 30 min. The solution was cooled to −78° C. again and Cy.sub.2PCl (0.25 mmol, 1 eq.) was added to the continuously stirred solution. The mixture was allowed to warm to room temperature and subsequently heated to 50° C. and kept at this temperature overnight. After the reaction was cooled down to RT, THF was removed and dried in vacuum, [Ir(COD)Cl].sub.2 (0.125 mmol) and DCM (5.0 mL) were added to the tube, stirred at 50° C. for 2 h. Then KPF.sub.6 (0.25 mmol) was added to the reaction mixture and stirred at RT for overnight. The reaction solution is rotary evaporated onto silica, loaded onto a column of silica prepared with DCM chromatographed with EtOAc/DCM: 1/10 to afford the orange solid after two times column chromatography (130 mg, 55%).
(81) .sup.1H-NMR (300 MHz, CD.sub.2Cl.sub.2) δ=7.37-7.02 (m, 3H), 5.74-5.56 (m, 1H), 5.52-5.46 (m, 1H), 4.47-4.30 (m, 1H), 3.45-3.21 (m, 1H), 3.19-2.92 (m, 3H), 2.66 (s, 3H), 2.63-2.48 (m, 2H), 2.44 (s, 3H), 2.40-2.19 (m, 2H), 2.16-1.70 (m, 15H), 1.68-1.46 (m, 6H), 1.41-1.28 (m, 13H), 1.18-0.95 (m, 5H), 0.71-0.58 (m, 1H). .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=118.42, −127.01, −132.85, −138.70, −144.55, −150.39, −156.24, −162.09. .sup.19F-NMR (376 MHz, CD.sub.2Cl.sub.2) δ=−72.64, −74.52. HR-MS (ESI) m/z calcd for C.sub.41H.sub.60NOPIr [M].sup.+806.4036 found 806.4061.
(82) Va-7
(83) The reaction was performed according to the above described standard procedure, using CgPBr (0.25 mmol, freshly prepared). The complex could be isolated as an orange solid (120 mg). There are two peaks showing in .sup.31P NMR. The HRMS shows that the desired complex is present.
(84) .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=89.41, 83.71. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ=−62.86. HR-MS (ESI) m/z calcd for C.sub.35H.sub.46NO.sub.4PIr [M].sup.+ 768.2788 found 768.2816.
(85) Va-8
(86) The reaction was performed according to the above described standard procedure, using 2-(tert-butylchlorophosphaneyl)pyridine (0.25 mmol). The complex could be isolated as a red solid (196 mg). There are 2 peaks showing in .sup.31P NMR at 107.59 and 102.97 ppm (1:1.4). The HRMS shows that the desired complex is present. .sup.31P-NMR (122 MHz, CD.sub.2Cl.sub.2) δ=107.59, 102.97. .sup.19F-NMR (282 MHz, CD.sub.2Cl.sub.2) δ=−62.83. HR-MS (ESI) m/z calcd for C.sub.34H.sub.43N.sub.2OPIr [M].sup.+ 719.2737 found 719.2757.
(87) Va-9
(88) A solution of the respective alcohol precursor in THF (0.25 mmol, in 5.0 mL THF) was cooled to −78° C. and n-BuLi (0.1 mL of a 2.5 M n-BuLi solution in hexane; 0.25 mmol; 1 eq.) was added dropwise to the continuously stirred solution. After completion of the addition the solution was allowed to warm to room temperature and was stirred at this temperature for further 30 min. The solution was cooled to −78° C. again and Cy.sub.2PCl (0.25 mmol, 1 eq.) was added to the continuously stirred solution. The mixture was allowed to warm to room temperature and subsequently heated to 50° C. and kept at this temperature overnight. After the reaction was cooled

down to RT, THF was removed and dried in vacuum, $[Ir(COD)Cl]_2$ (0.125 mmol) and DCM (5.0 mL) were added to the tube, stirred at 50° C. for 2 h. Then $Li\{A1[OC(CF_3)_3]_4\}$ (0.275 mmol) was added to the reaction mixture and stirred at RT for overnight. The reaction solution is rotary evaporated onto silica, loaded onto a column of silica prepared with DCM chromatographed with n-heptane/DCM: 1/1 to afford the orange solid (122 mg, 28%).

(89) ^1H-NMR (300 MHz, CD_2Cl_2) δ=7.40-7.12 (m, 4H), 5.67-5.63 (m, 1H), 5.50-5.42 (m, 1H), 4.43-4.36 (m, 1H), 3.39-3.30 (m, 1H), 3.25-2.90 (m, 3H), 2.67 (s, 3H), 2.60-2.42 (m, 4H), 2.42-2.18 (m, 2H), 2.13-1.99 (m, 5H), 1.96-1.71 (m, 9H), 1.65-1.52 (m, 5H), 1.45-1.01 (m, 12H), 0.69-0.53 (m, 1H). ^{31}P-NMR (121 MHz, CD_2Cl_2) δ=118.81. ^{19}F-NMR (282 MHz, CD_2Cl_2) δ=−75.74. HR-MS (ESI) m/z calcd for $C_{37}H_{52}NOPIr$ $[M]^+$ 750.3416 found 750.3420.

(90) Va-10 The reaction was performed according to the above described procedure using 287 mg of $[Ir(COD)_2]BARF$ (0.225 mmol). The complex could be isolated as an orange solid (148 mg; 40% based on $[Ir(COD)_2]BARF$).

(91) ^1H-NMR (300 MHz, CD_2Cl_2): δ (ppm)=7.91-7.46 (m, 12H), 7.21 (s, 1H), 7.09 (s, 1H), 6.94 (s, 1H), 5.67-5.63 (m, 1H), 5.46-5.41 (m, 1H), 4.38-4.36 (m, 1H), 3.36-3.32 (m, 1H), 3.19-2.85 (m, 3H), 2.64 (s, 3H), 2.53-2.46 (m, 2H), 2.41 (s, 3H), 2.35 (s, 3H), 2.31-2.18 (m, 2H), 2.19-1.83 (m, 14H), 1.68-1.54 (m, 6H), 1.38-1.20 (m, 5H), 1.14-0.97 (m, 5H), 0.68-0.56 (m, 1H). ^{31}P-NMR (122 MHz, CD_2Cl_2) δ=118.64. ^{19}F-NMR (282 MHz, CD_2Cl_2) δ=−62.87. HR-MS (ESI) m/z calcd for $C_{38}H_{54}NOPIr$ $[M]^+$ 764.3572 found 764.3577.

(92) Va-11

(93) The reaction was performed according to the above described procedure. The complex could be isolated as an orange solid (274 mg; 73% based on $[Ir(COD)_2]BARF$).

(94) ^1H-NMR (300 MHz, CD_2Cl_2): δ (ppm)=7.79-7.66 (m, 8H), 7.56 (s, 4H), 7.29 (s, 1H), 7.23 (s, 1H), 7.13 (s, 1H), 5.65 (td, J=5.9, 2.2 Hz, 1H), 5.46-5.40 (m, 1H), 4.42-4.36 (m, 1H), 3.38-3.30 (m, 1H), 3.19-2.86 (m, 3H), 2.65 (s, 3H), 2.59-2.44 (m, 2H), 2.42 (s, 3H), 2.38-1.54 (m, 20H), 1.46-0.98 (m, 21H), 0.70-0.58 (m, 1H). ^{31}P-NMR (122 MHz, CD_2Cl_2) δ (ppm)=118.67. ^{19}F-NMR (282 MHz, CD_2Cl_2) δ (ppm)=−62.86. HR-MS (ESI) m/z calcd for $C_{41}H_{60}NOPIr$ $[M]^+$ 806.4042 found 806.4053.

(95) Va-12

(96) The reaction was performed according to the above described procedure using 287 mg of $[Ir(COD)_2]BARF$ (0.225 mmol). The complex could be isolated as an orange solid (298 mg; 82% based on $[Ir(COD)_2]BARF$).

(97) ^1H-NMR (300 MHz, CD_2Cl_2): δ (ppm)=7.80-7.52 (m, 12H), 7.42-7.19 (m, 3H), 7.12 (d, J=7.5 Hz, 1H), 5.65 (td, J=5.6, 2.6 Hz, 1H), 5.48-5.42 (m, 1H), 4.43-4.37 (m, 1H), 3.38-3.30 (m, 1H), 3.21-2.89 (m, 3H), 2.67 (s, 3H), 2.58-2.45 (m, 2H), 2.42 (s, 3H), 2.38-2.16 (m, 2H), 2.13-2.05 (m, 3H), 2.02-1.89 (m, 4H), 1.84 (s, 3H), 1.81-1.72 (m, 2H), 1.64-1.49 (m, 3H), 1.39-1.19 (m, 8H), 1.12-0.99 (m, 4H), 0.68-0.56 (m, 1H). ^{31}P-NMR (122 MHz, CD_2Cl_2) δ=118.80. ^{19}F-NMR (282 MHz, CD_2Cl_2) δ=−62.88. HR-MS (ESI) m/z calcd for $C_{37}H_{52}NOPIr$ $[M]^+$ 750.3416 found 750.3420.

EXAMPLES

(98) Reactions were performed in metal autoclaves. Reaction mixtures were analyzed without workup via HPLC (Chiralpak IC column, 95/5 heptane/ethanol, 1 mL/min) or SFC (OZ-H column, 2.5% MeOH in supercritical CO_2, 3 mL/min) chromatography.

Examples 1-12

(99) The Ir-complex (catalyst loading given) and 0.64 g 1-(2,2,4-trimethyl-1-quinolyl)ethanone (3 mmol) were placed in an 8-mL autoclave vial containing a PTFE-coated stirring bar. The autoclave vial was closed using a screw cap with septum and flushed with argon (10 min). Hexafluoroisopropanol (HFIP, 4 mL) was added via the septum to the vial. The vial was placed in

an argon containing autoclave and the autoclave was flushed with argon (10 min). The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for the given time. After cooling to room temperature and depressurizing, the vial was taken out of the autoclave and the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis. Typical values are given.

(100) TABLE-US-00003 TABLE 2 catalyst Reaction loading Conversion Enantiomeric Example Catalyst time (h) (mol %) GC (% a/a) excess (% ee) 1 Va-6 3 0.02 78.8 98.8 2 Va-11 3 0.02 85.2 98.8 3 Va-4 3 0.02 <1 — 4 Va-5 3 0.02 58 n.d. 5 Va-7 3 0.02 <1 — 6 Va-8 3 0.02 2 — 7 Va-3 3 0.02 37.5 n.d. 8 Va-1 16 0.025 98.2 96.2 9 Va-1 6 0.03 94.9 n.d. 10 Va-2 6 0.03 96.1 n.d. 11 Va-9 16 0.03 81.3 n.d. 12 Va-12 16 0.03 79.4 n.d.

Examples 13-18

(101) The Ir-complex (catalyst loading given) and 2.56 g 1-(2,2,4-trimethyl-1-quinolyl)ethanone (12 mmol) were placed in an 25-mL autoclave. The autoclave was flushed with argon (10 min). Hexafluoroisopropanol (HFIP, 16 mL) was added to the autoclave. The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for the given time. After cooling to room temperature and depressurizing, the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis.

(102) TABLE-US-00004 TABLE 3 catalyst Reaction loading Conversion Example Catalyst time (h) (mol %) GC (% a/a) 13 Va-10 17 0.01 82.5 14 Va-10 40 0.01 93.9 15 Va-1 17 0.01 93.1 16 Va-1 40 0.01 98.0 17 Va-1 6 0.01 50.8 18 Va-2 6 0.01 53.4

Examples 19-48

(103) The Ir-complex Va-1 (catalyst loading given) and 0.64 g 1-(2,2,4-trimethyl-1-quinolyl)ethanone (3 mmol, purified with heptane: water wash+crystallization) were placed in an 8-mL autoclave vial containing a PTFE-coated stirring bar. The autoclave vial was closed using a screw cap with septum and flushed with argon (10 mi). Hexafluoroisopropanol (HFIP, 4 mL) and additive (loading given) were added via the septum to the vial. The vial was placed in an argon containing autoclave and the autoclave was flushed with argon (10 min). The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for the given time. After cooling to room temperature and depressurizing, the vial was taken out of the autoclave and the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis. Typical values are given.

(104) TABLE-US-00005 TABLE 4 catalyst Additive Reaction loading Conversion Enantiomeric Example (mol %) time (h) (mol %) GC (% a/a) excess (% ee) 19 — 16 0.02 95.3 n.d. 20 — 21 0.02 95.5 n.d. 21 — 3 0.02 55.2 n.d. 22 — 16 0.03 97.6 n.d. 23 Pentafluorophenol (1) 16 0.02 97.2 n.d. 24 1,2,2,6,6-Pentamethylpiperidin (1) 16 0.02 67.1 n.d. 25 Nonafluoro-tert-butyl alcohol (1) 16 0.03 96.3 n.d. 26 Nonafluoro-tert-butyl alcohol (5) 16 0.03 97.5 n.d. 27 3,5-bis-trifluorophenol (1) 16 0.02 95.7 n.d. 28 AcOH (1) 16 0.02 96 n.d. 29 AcOH (5) 3 0.02 66.5 n.d. 30 AcOH (10) 3 0.02 63.7 n.d. 31 AcOH (20) 3 0.02 54.2 n.d. 32 HPF.sub.6 (1) 3 0.02 >99 n.d. 33 HBF.sub.4*OEt.sub.2 (1) 16 0.02 90.5 n.d. 34 TfOH (1) 16 0.02 76.9 n.d. 35 Sc(OTf).sub.3 (1) 3 0.02 >99 99 36 BF.sub.3*OEt.sub.2 (1) 3 0.02 98.9 98 37 BH.sub.3*THF (1) 3 0.02 69.8 n.d. 38 BF.sub.3*AcOH (1) 3 0.02 >99 n.d. 39 BF.sub.3*n-PrOH (1) 3 0.02 >99 n.d. 40 Al(OTf).sub.3 (1) 3 0.02 >99 n.d. 41 AlF.sub.3 (1) 3 0.02 65.9 n.d. 42 AlMe.sub.3 (1) 3 0.02 91.1 n.d. 43 Ti(O.sup.iPr).sub.4 (1) 3 0.02 90.7 n.d. 44 BPh.sub.3 (1) 3 0.02 85.4 n.d. 45 B(C.sub.6F.sub.5).sub.3 (1) 3 0.02 >99 .sup. 97.6 46 B(C.sub.6F.sub.5).sub.3 (0.5) 3 0.02 97.3 n.d. 47 B(C.sub.6F.sub.5).sub.3 (0.1) 3 0.02 63.3

n.d. 48 B(OH).sub.3 (1) 3 0.02 72.7 n.d.

Examples 49-54

(105) The Jr-complex Va-1 (catalyst loading given) and 1-(2,2,4-trimethyl-1-quinolyl)ethanone (amount given; purified with heptane: water wash+crystallization) were placed in an 25-mL autoclave. The autoclave was flushed with argon (10 min). Hexafluoroisopropanol (1.33 mL per mmol of 1-(2,2,4-trimethyl-1-quinolyl)ethenone)) and additive (loading given) were added to the autoclave. The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for the given time. After cooling to room temperature and depressurizing, the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis.

(106) TABLE-US-00006 TABLE 5 catalyst Additive Scale (amount of Reaction loading Conversion Enantiomeric Example (mol %) compound (II)) time (h) (mol %) GC (% a/a) excess (% ee) 49 — 9 mmol 6 0.01 50.8 n.d. 50 B(C.sub.6F.sub.5).sub.3 (0.5) 9 mmol 20 0.01 85.2 n.d. 51 BF.sub.3*OEt.sub.2 (1) 10 mmol   16 0.01 99.2 n.d. 52 Al(OTf).sub.3 (1) 10 mmol   16 0.01 >99 n.d. 53 HPF.sub.6 (1) 9 mmol 16 0.01 97.3 n.d. 54 BF.sub.3*AcOH (1) 9 mmol 16 0.01 98.1 n.d.

Examples 55-56

(107) The Ir-complex (identifier and catalyst loading given) and 0.64 g 1-(2,2,4-trimethyl-1-quinolyl)ethanone (3 mmol, purified with heptane: water wash+crystallization) were placed in an 8-mL autoclave vial containing a PTFE-coated stirring bar. The autoclave vial was closed using a screw cap with septum and flushed with argon (10 min). Hexafluoroisopropanol (HFIP, 4 mL) and BF.sub.3*OEt.sub.2 (1 mol % with respect to 1-(2,2,4-trimethyl-1-quinolyl)ethanone) were added via the septum to the vial. The vial was placed in an argon containing autoclave and the autoclave was flushed with argon (10 min). The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for the given time. After cooling to room temperature and depressurizing, the vial was taken out of the autoclave and the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis. Typical values are given.

(108) TABLE-US-00007 TABLE 6 catalyst Additive Reaction loading Conversion Enantiomeric Example Catalyst (mol %) time (h) (mol %) GC (% a/a) excess (% ee) 55 Va-6 — 3 0.02 78.8 98.8 56 Va-6 BF.sub.3*OEt.sub.2 (1) 3 0.02 94.2 99

Examples 57-60

(109) The Ir-complex Va-1 (0.02 mol %, 0.6 mol) and 0.64 g 1-(2,2,4-trimethyl-1-quinolyl)ethanone (3 mmol, purified with heptane: water wash+crystallization) were placed in an 8-mL autoclave vial containing a PTFE-coated stirring bar. The autoclave vial was closed using a screw cap with septum and flushed with argon (10 min). 2,2,2-Trifluoroethanol (TFE, 4 mL) and BF.sub.3*OEt.sub.2 (loading given) were added via the septum to the vial. The vial was placed in an argon containing autoclave and the autoclave was flushed with argon (10 min). The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for 3 h. After cooling to room temperature and depressurizing, the vial was taken out of the autoclave and the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis. Typical values are given.

(110) TABLE-US-00008 TABLE 7 BF.sub.3*OEt.sub.2 Conversion Example (mol %) GC (% a/a) 57 — <1 58 1 86 59 3 88 60 5 82

Comparative Examples

(111) The Ir-complex (catalyst loading given) and 0.64 g 1-(2,2,4-trimethyl-1-quinolyl)ethanone (3 mmol) were placed in an 8-mL autoclave vial containing a PTFE-coated stirring bar. The autoclave vial was closed using a screw cap with septum and flushed with argon (10 min). Hexafluoroisopropanol (HFIP, 4 mL) was added via the septum to the vial. The vial was placed in an argon containing autoclave and the autoclave was flushed with argon (10 min). The autoclave was pressurized with hydrogen gas (10 bar) and subsequently depressurized to atmospheric pressure three times. After this the autoclave was pressurized to 60 bar hydrogen pressure and was placed in a suitable alumina block. After heating to 85° C. the reaction was kept at this temperature for the given time. After cooling to room temperature and depressurizing, the vial was taken out of the autoclave and the reactions outcome was determined by GC-FID analysis (deluted with EtOH) and the enantiomeric excess by HPLC analysis.
(112) TABLE-US-00009 TABLE 8 Catalyst loading Example Catalyst (mol %) Time (h) Conversion 17 Va-1 0.01 6 50.8 18 Va-2 0.01 6 53.4 15 Va-1 0.01 17 93.1 13 Va-10 0.01 17 82.5 61 Va-10 0.025 16.5 98 62 Va-13 0.1 16 99.2 63 Ir catalyst (1) 0.1 16 84
(113) This collection of experimental results shows the superiority of complexes Va-1 and Va-2. Va-2 performs gives slightly higher conversion than Va-1 after 6 h with 0.01 mol % of catalyst. Va-1 gives significantly higher conversion than Va-10 (=Va-10 from WO2019/185541 A1) after 17 h with 0.01 mol % of catalyst. Va-10 gives similar performance than Va-13 (=Va-1 from WO2019/185541 A1) although only ¼ of the catalyst amount was used. It must thus be considered superior to Va-13. Va-13 gives superior conversion compared to Ir catalyst (I) from DE112015001290 T5. Moreover, all of the catalyst Va gives superior enantioslectivities (>95% ee) compared to Ir catalyst (1) from DE112015001290 T5 (81.7% ee). Catalyst Va-1 gives higher conversion (93.1 vs 84%) than Ir catalyst (1) from DE112015001290 T5 even at only one-tenths ofthe catalyst loading (0.01 vs 0.1 mol %).
(114) ##STR00016##

Claims

1. A process for preparing a compound of formula (Ia) or (Ib), ##STR00017## wherein R.sup.1 is selected from the group consisting of C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.1-C.sub.6-alkoxy-C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl, C.sub.6-C.sub.14-aryl, or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein the C.sub.1-C.sub.6-alkyl, C.sub.3-C.sub.6-cycloalkyl and the C.sub.1-C.sub.6-alkoxy in the C.sub.1-C.sub.6-alkoxy-C.sub.1-C.sub.6-alkyl moiety, are optionally substituted by 1 to 3 substituents independently selected from the group consisting of halogen, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-haloalkoxy and phenyl, wherein the phenyl may be substituted by one to five substituents selected independently from each other from halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkyl, and C.sub.1-C.sub.4-haloalkoxy, and wherein the C.sub.6-C.sub.14-aryl and the C.sub.6-C.sub.14-aryl in the C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl moiety in each case is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, R.sup.2 and R.sup.3 are the same and are selected from the group consisting of hydrogen, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl and C.sub.1-C.sub.6-alkoxy-C.sub.1-C.sub.6-alkyl, or R.sup.2 and R.sup.3 together with the carbon which they are bound to, form a C.sub.3-C.sub.6-cycloalkyl ring, R.sup.4 is hydrogen, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.1-C.sub.6-alkoxy, C.sub.1-C.sub.6-haloalkoxy, C.sub.1-C.sub.6-alkylamino, C.sub.2-C.sub.6-alkenyl, C.sub.2-C.sub.6-alkynyl, C.sub.3-C.sub.6-cycloalkyl, C.sub.3-C.sub.6-cycloalkyl-C.sub.1-C.sub.4-alkyl, C.sub.2-C.sub.6-alkenyloxy, 9-flurorenylmethyleneoxy, C.sub.6-C.sub.14-aryl, C.sub.6-C.sub.14-aryloxy, C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyloxy or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein the C.sub.6-C.sub.14-aryl as such or as part

of a composite substituent is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, n is 0, 1, 2, 3 or 4, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl, C.sub.1-C.sub.6-haloalkyl, C.sub.1-C.sub.6-alkoxy, hydroxyl, amino and —C(═O)—C.sub.1-C.sub.6-alkyl, the process comprising enantioselective hydrogenation of a compound of formula (II) ##STR00018## wherein the substituents R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5 and the integer n are each as defined for the compound of formula (Ia) or (Ib), in presence of a chiral iridium catalyst, wherein the chiral iridium catalyst is of the general formula (Va) or (Vb) ##STR00019## wherein R.sup.6 is 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl, Y is a non-coordinating anion selected from the group consisting of [B(R.sup.18).sub.4].sup.− and [Al{OC(CF.sub.3).sub.3}.sub.4].sup.−of formula (VII), ##STR00020## wherein R.sup.18 is 3,5-bis(trifluoromethyl)phenyl.
2. The process according to claim 1, wherein R.sup.1 is C.sub.1-C.sub.6-alkyl or C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl, wherein C.sub.6-C.sub.14-aryl in the C.sub.6-C.sub.14-aryl-C.sub.1-C.sub.4-alkyl moiety is unsubstituted or substituted by one to five substituents selected from the group consisting of halogen, C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy and C.sub.1-C.sub.4-haloalkoxy, R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkoxy, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl.
3. The process according to claim 1, wherein R.sup.1 is C.sub.1-C.sub.6-alkyl, R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, C.sub.1-C.sub.4-alkoxy, C.sub.1-C.sub.4-haloalkoxy, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen, C.sub.1-C.sub.6-alkyl and C.sub.1-C.sub.6-haloalkyl.
4. The process according to claim 1, wherein R.sup.1 is C.sub.1-C.sub.6-alkyl, R.sup.2 and R.sup.3 are the same and are selected from C.sub.1-C.sub.4-alkyl, or R.sup.2 and R.sup.3 together with the carbon which they are bound to, form a C.sub.3-C.sub.6-cycloalkyl ring, R.sup.4 is C.sub.1-C.sub.4-alkyl, C.sub.1-C.sub.4-haloalkyl, phenyl or benzyl, n is 0, 1 or 2, each substituent R.sup.5, if present, is independently selected from the group consisting of halogen and C.sub.1-C.sub.6-alkyl.
5. The process according to claim 1, wherein R.sup.1 is C.sub.1-C.sub.4-alkyl, R.sup.2 and R.sup.3 are methyl, R.sup.4 is C.sub.1-C.sub.4-alkyl, n is 0 or 1, R.sup.5 if present, is fluorine.
6. The process according to claim 1, wherein the process is performed in presence of an additive, which is selected from the group consisting of Bronsted acids, Lewis acids, and mixtures thereof.
7. The process according to claim 6, wherein the additive is selected from the group consisting of hexafluorophosphoric acid, pentafluorophenol, 3,5-bis(trifluoromethyl)phenol, triphenylborane, tris[3,5-bis(trifluoro-methyl)phenyl]borane, tris(2,3,4,5,6-pentafluorophenyl)borane, aluminum (III) trifluoromethanesulfonate, scandium (III) trifluoromethanesulfonate, aluminum (III) fluoride, titanium (IV) isopropoxide, trimethyl aluminum, boron trifluoride, complexes of boron trifluoride, and mixtures thereof.
8. The process according to claim 1, wherein the hydrogenation is conducted using hydrogen gas at a pressure of from 1 to 300 bar.
9. The process according to claim 1, wherein the amount of chiral iridium catalyst used is within the range of from 0.001 mol % to 5 mol %, based on the amount of the compound of formula (II).
10. The process according to claim 1, wherein the hydrogenation is conducted at a temperature within the range of from 20° C. to 130° C.
11. The process according to claim 1, wherein the amount of additive used is within the range of

from 0.1 mol % to 10 mol %.

12. A chiral iridium catalyst selected from chiral iridium catalysts of general formulae (Va) and (Vb) ##STR00021## wherein R.sup.6 is 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl, Y is a non-coordinating anion selected from the group consisting of [B(R.sup.18).sub.4].sup.− and [Al{OC(CF.sub.3).sub.3}.sub.4].sup.− of formula (VII), ##STR00022## wherein R.sup.18 is 3,5-bis(trifluoromethyl)phenyl.

13. The chiral iridium catalyst according to claim 12, wherein the chiral iridium catalyst is the chiral iridium catalyst of formula (Va) ##STR00023## wherein R.sup.6 is 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl, Y is the non-coordinating anion [B(R.sup.18).sub.4].sup.−, wherein R.sup.18 is 3,5-bis(trifluoromethyl)phenyl.

14. The chiral iridium catalyst according to claim 12, wherein the chiral iridium catalyst is the chiral iridium catalyst of formula (Va) ##STR00024## wherein R.sup.6 is 2,6-diethyl-4-methylphenyl, R.sup.7 is hydrogen, R.sup.8 is methyl, R.sup.9 and R.sup.10 are both cyclohexyl, Y is the non-coordinating anion [Al{OC(CF.sub.3).sub.3}.sub.4].sup.−of formula (VII) ##STR00025##